

Software Requirements Specification (SRS)

Students: Ana Júlia Pereira Silva and Lara Abreu Costa de Freitas

Professor: Daniel Linhares Lim Apo

Sumário

1. Introduction	2
2. Business Objectives:	2
3. Stakeholders.....	3
4. Business Requirements	3
5. Screens.....	4
5.1. Sign Up Screen.....	4
5.2. Login and Password Recovery Screen	7
5.3. Fanfiction Addition/Cataloging Screen (Web Scraping)	8
5.4. Search/Explore Screen	10
5.5. Fanfic Details Screen (Rating and Review.....	11
5.6. Shelves Screen (My Library)	13
5.7. User Profile Screen.....	14
5.8. Feed and Social Screen (Community).....	15
6. Global / Cross-Cutting Requirements (Infrastructure)	17
7. Process Flow (Flowchart)	18
8. Entity-Relationship Model.....	19
9. Success Criteria	19
Adoption and Retention Criteria:.....	20
Operational Efficiency Criteria (Automation):	20
Community Engagement Criteria:	20
Performance and Technical Quality Criteria:.....	20

1. Introduction

- **Purpose of the document:** This document aims to define the business needs, scope, and assumptions for the development of the FOS (Fics On Shelf) platform, ensuring that all stakeholders understand the purpose and value of the solution.
- **Project Scope:** FOS is a web application focused on cataloging, managing, and fostering social interaction around the reading of fanfictions. The scope covers everything from automated data capture of works on external platforms (AO3, Wattpad) to the creation of a social ecosystem where readers can rate, review, and share their readings.
 - **Business problem or opportunity to be solved:** Currently, avid fanfiction readers consume works scattered across various platforms (AO3, Wattpad, Spirit, etc.). This creates a dispersion problem: readers lose track of what they have already read, where they left off, which fics were their favorites, and lack a centralized place to organize their virtual shelves or interact with other readers about these specific works. FOS emerges as the opportunity to be the “Goodreads/Skoob” exclusively for the fanfiction niche.

2. Business Objectives:

- **Centralization and Organization:** Provide a unique environment where the user can concentrate and manage their entire fanfiction literary life, regardless of the source site of the work.
- **Increase Reader Productivity:** Reduce the time spent manually filling out reading records through automation (Web Scraping), which populates the work’s data simply by using the link.
- **Engagement and Community:** Foster an engaged community through social features (following friends, liking reviews, activity feed), increasing user retention on the platform.

3. Stakeholders

Stakeholder	Role / Interests	Responsibilities
Readers	End users / Final consumers. Seek organization, ease of use, and social interaction.	Provide reading data, evaluate works, and interact with the community.
Administrator/ Product Owner	Product owner. Seeks user growth, platform stability, and engagement.	Define scope, prioritize features, and maintain product vision.
Source Platforms (AO3, Wattpad)	Passive stakeholders. They are the source from which the primary data of the works is extracted.	None (FOS acts as an external aggregator).

4. Business Requirements

Business Requirement	Description

RB01	The system must allow centralized cataloging of fanfictions from multiple internet platforms.
RB02	The system must automate data capture (title, author, cover) to minimize user effort when adding a new reading
RB03	The company/platform must offer a system to control reading status (Read, Reading, Want to Read, Abandoned).
RB04	The system must enable social interaction among users, allowing the sharing of evaluations and critical reviews of works.
RB05	The platform must ensure privacy options, allowing users to keep their reading habits hidden if they wish.

5. Screens

5.1. Sign Up Screen

Functional Requirements

Code	Description
RF01	The system must allow the user to sign up by providing Email, Username, Display Name, and Password.
RF02	The system must validate the uniqueness of the email and username.
RF03	The system must require a password with at least 8 characters, including letters and numbers.
RF04	The system must send a confirmation email to validate the user's account.
RF05	The system must provide links to the "Terms of Use" and "Privacy Policy" and require the user to check a mandatory acceptance box before completing registration.

Non-Functional Requirements

Code	Description
RNF01	The system must not store the password in plain text. It must be converted into a hash (BCrypt) (LGPD/GDPR).
RNF02	The account creation and database persistence process must take less than 2 seconds.
RNF03	The sign-up service must be available 99% of the time.
RNF04	The system must record the date, time, and version of the Terms of Use accepted by the user at the time of registration, ensuring compliance with LGPD/GDPR (Legal audit).

UC01 – Cadastro no Sistema

UC01	System Registration
Objective	Create an account in the FOS system.
Actor:	New user
Main Flow	1. User enters Email, Username, Display Name, and Password. 2. User reads and checks the box agreeing to the Terms of Use and Privacy Policy. 3. System validates the provided data. 4. System creates the account and saves it in the database, recording acceptance of the terms. 5. System sends a confirmation email. 6. User is redirected to the login screen.
Alternative Flow	• If the user tries to register without checking the Terms of Use box → the system disables the “Create Account” button or displays a mandatory acceptance alert. • If email or username is already registered → the system displays an error message. • If the password does not meet the 8-character, letters, and numbers requirement → the system requests a new password.

5.2. Login and Password Recovery Screen

Functional Requirements

Code	Description
RF06	The system must allow login using the registered Email or Username + Password.
RF07	The system must return a generic error (“Invalid username or password”) in case of failure.
RF08	The system must allow the user to stay logged in (“Remember me”).
RF09	The system must provide the “Forgot my password” functionality via email.

Non-Functional Requirements

Code	Description
RNF05	The system must use JWT to manage the session securely and in a stateless manner.
RNF06	After 5 failed attempts in less than 1 minute, the system must block the IP for 15 minutes.

UC02 – Authentication and Recovery

UC02	Detalhamento
Objective	Authenticate the user to access the platform or recover access.
Actor	Registered user.

Main Flow	• User enters credentials (Email/Username and Password). - User may check the “Remember me” option. - System validates credentials, generates the JWT token, and grants access. - User is redirected to their Main Shelf.
Alternative Flow	- If incorrect data is entered → system displays generic error (“Invalid username or password”). - If 5 errors occur within 1 minute → system blocks the user’s IP for 15 minutes. - If the user clicks “Forgot my password” → system requests the email and sends a recovery link.

5.3. Fanfiction Addition/Cataloging Screen (Web Scraping)

Functional Requirements

Code	Description
RF10	The system must allow pasting a URL from supported platforms.
RF11	The system must check if the URL already exists in the database; if so, redirect the user.
RF12	The system must attempt to automatically extract Title, Author, Tags, and Cover.

RF13	If extraction fails, the system must open a form for manual input.
RF14	The system must identify and automatically save the source platform.
RF15	The system must extract the original cover or generate a default cover, allowing changes via upload.

Non-Functional Requirements

Code	Description
RNF07	Covers must be optimized/compressed for fast loading on 3G/4G.

UC03 – Work Cataloging

UC03	Detail
Objective	Add a fanfiction to the database and to the reader's shelf.
Actor	Logged-in user.
Main Flow	1. User pastes the fanfiction link. 2. System identifies the source platform. 3. System performs scraping (extracts title, author, tags, cover). 4. User chooses which shelf to save it to (e.g., "Reading"). 5. Work is saved and added to the profile

Alternative Flow	• If the link already exists in the database → system skips scraping and redirects to the existing fanfic page. • If the system cannot read the link data → opens a blank form for manual input. • If the fanfiction has no cover → system applies a default image
-------------------------	--

5.4. Search/Explore Screen

Functional Requirements

Code	Description
RF16	User searches by keywords (Title or Author).
RF17	The system allows filtering results by specific platform.
RF18	User can search fanfics by Tags (e.g., “Enemies to Lovers”).
RF19	If no results are found, display the button: “Be the first to add this fanfiction!”.

Non-Functional Requirements

Code	Description

UC04 – Fanfic Search

UC04	Detail
Objective	Find new stories or check if a fanfiction is already in FOS.
Actor	Logged-in user.
Main Flow	1. User enters keywords in the search bar. 2. Applies desired filters (Platform or Tags). 3. System searches the database and displays the list of corresponding fanfics
Alternative Flow	• If no works are found → system displays an empty screen with the button “Didn’t find it? Be the first to add this fanfiction!”

5.5. Fanfic Details Screen (Rating and Review

Functional Requirements

Code	Description
RF20	User can rate from 1 to 5 stars.
RF21	User can write a text comment (review).
RF22	The system must automatically record the date of the review.
RF23	Option to mark “Contains Spoiler,” hiding the text from others.

RF24	System calculates and displays the arithmetic average of ratings in real time.
RF25	System generates a link/image for external sharing of the review.

Non-Functional Requirements

Code	Description

UC05 – Reading Review

UC05	Detail
Objective	Record the user's opinion and rating of a work.
Actor	Logged-in user.
Main Flow	1. User accesses the fanfiction page. 2. Selects the number of stars (1 to 5) and writes their review. 3. System saves the review, recording the current date. 4. System recalculates the global average rating of the work.
Alternative Flow	• If the review contains story spoilers → user checks the "Contains Spoiler" box → system covers the text with an alert requiring a click to read.

5.6. Shelves Screen (My Library)

Functional Requirements

Code	Description
RF26	The system offers 5 categories: Read, Reading, Want to Read, Abandoned, Favorite.
RF27	User marks reading progress (e.g., “Chapter 15 of 40”).
RF28	Allows creation of personalized lists.
RF29	User defines the shelf as “Public” or “Private”.

Non-Functional Requirements

Code	Description
RNF08	User must have full control over who can view their lists (privacy).

UC06 – Reading Organization

UC06	Detail
Objective	Organize and monitor the user’s reading status.
Actor	Logged-in user.

Main Flow	1. User accesses their personal library. 2. Moves a fanfiction to the desired category (e.g., from “Want to Read” to “Reading”). 3. Updates the chapter they stopped at. 4. System saves changes instantly.
Alternative Flow	• If the user wants extra organization → clicks “New List,” names the custom list, and adds fanfiction to it. • If privacy is set to “Private” → other users cannot see the fanfictions in that list on the profile

5.7. User Profile Screen

Functional Requirements

Code	Description
RF30	User adds avatar, banner, and edits bio.
RF31	Profile highlights the “Top 4” favorites (showcase).
RF32	System displays reading metrics (Total read, Most read tags, etc.).
RF33	User configures whether the profile is public or private.

Non-Functional Requirements

Code	Description
------	-------------

--	--

UC07 – Profile Customization

UC07	Detail
Objective	Customize the reader's visual identity and data within the Community.
Actor	Logged-in user.
Main Flow	2. • User accesses profile settings. 3. Uploads images (Avatar/Banner) and writes a biography. 4. Selects 4 fanfics from their history to feature in the "Top 4." 5. System consolidates and displays aesthetic reading statistics (charts).
Alternative Flow	• Se a conta for definida como "Privada" → sistema oculta estatísticas e vitrine de quem não for um seguidor aprovado.

5.8. Feed and Social Screen (Community)

Functional Requirements

Code	Description
RF34	The user can follow other profiles to keep track of activities.
RF35	The Feed shows recent actions of the people the user follows.

RF36	The system allows liking and commenting on other users' reviews.
------	--

Non-Functional Requirements

Code	Description

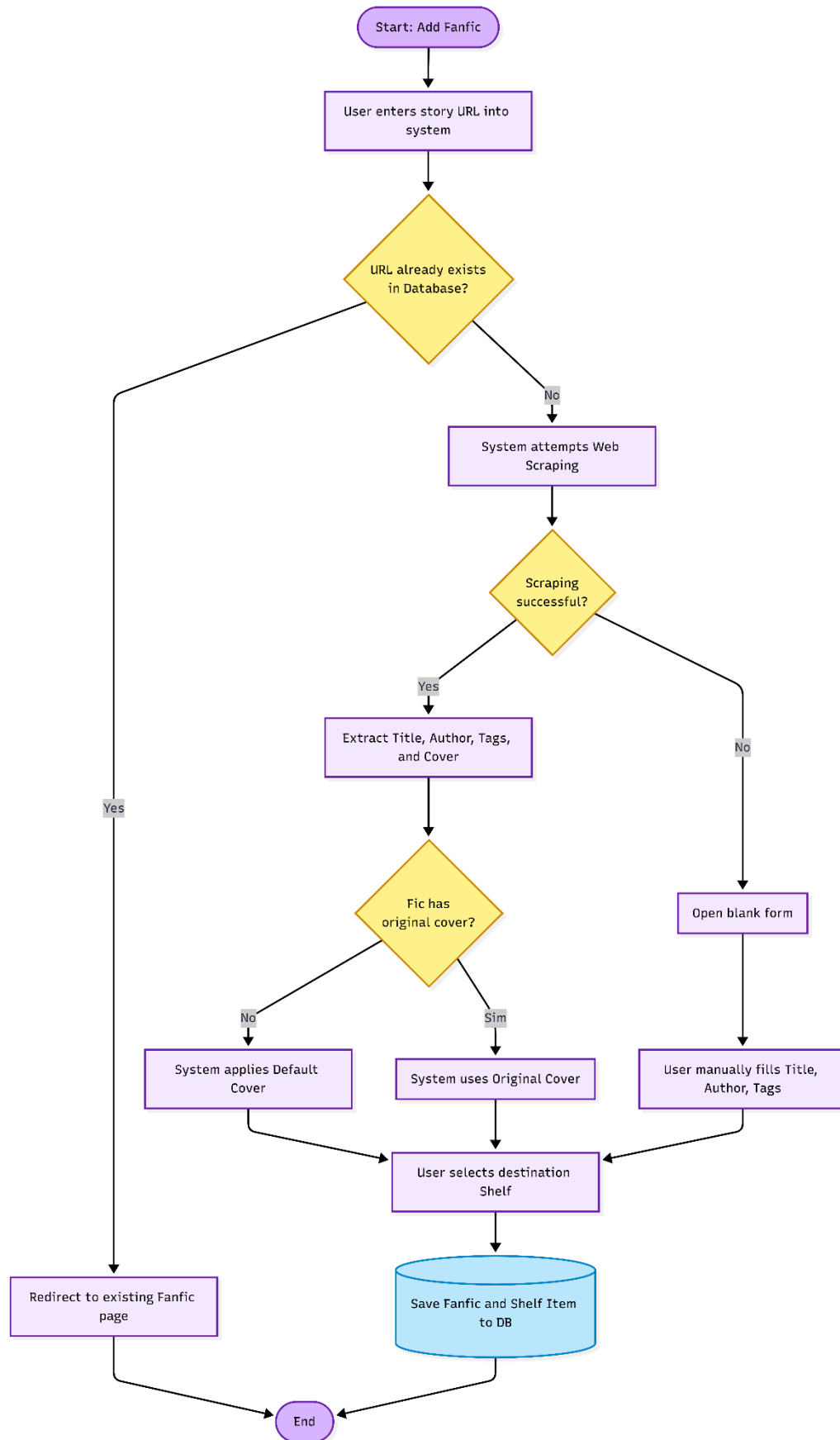
UC08 – Avaliação de Leitura

UC08	Detail
Objective	Engage with other readers and discover new readings through friends.
Actor	Logged-in user.
Main Flow	1. User opens the “Feed” page. 2. System loads recent updates from the profiles they follow. 3. User reads a review in the feed and clicks “Like” or “Comment.” 4. The review’s author receives the interaction.
Alternative Flow	• If the account is set to “Private” → system hides statistics and showcase from non-approved followers.

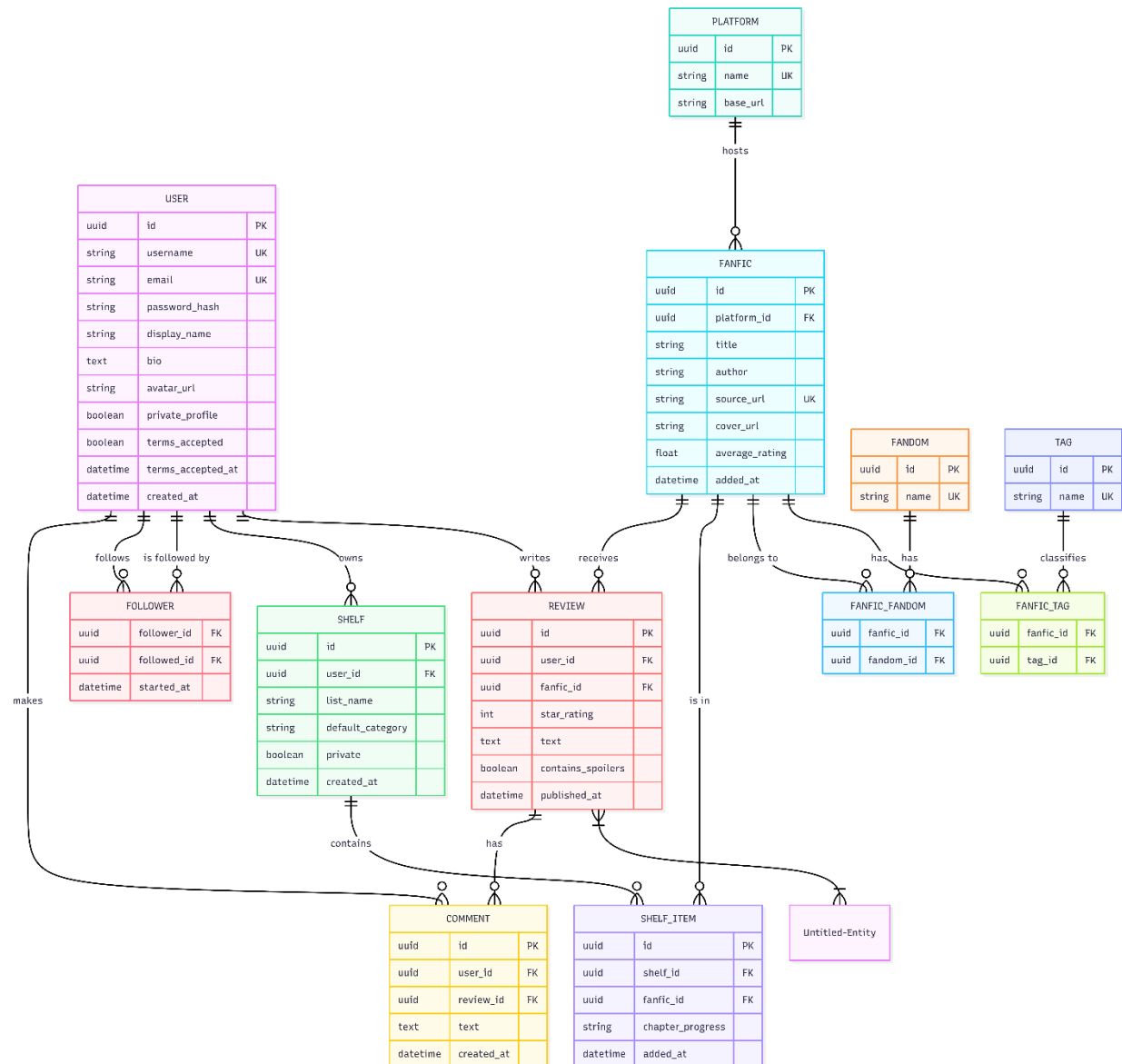
6. Global / Cross-Cutting Requirements (Infrastructure)

ID	Type	Requirement Description
RNF.09	Security	All data exchange between front-end and back-end must occur via HTTPS protocol.
RNF.10	Scalability	The Database must support growth from 100 to 100,000 users.
RNF.11	UX/UI	Mobile-First interface, ensuring perfect functionality on smartphones.
RNF.12	UX/UI	<i>Dark Mode (Purple Theme) set as the system default.</i>

7. Process Flow (Flowchart)



8. Entity-Relationship Model



9. Success Criteria

To ensure that the FOS (Fics On Shelf) solution meets business needs and resolves the dispersion problem in fanfiction organization, the project will be considered successful if it achieves the following indicators within 3 months of launch:

Adoption and Retention Criteria:

- Reach 500 registered users organically.
- Maintain a retention rate where at least 40% of registered users access the platform weekly (Weekly Active Users).

Operational Efficiency Criteria (Automation):

- The Web Scraping module (automatic capture via link) must achieve a success rate equal to or greater than 90% when importing links from supported platforms, reducing manual input time for the user to less than 5 seconds.

Community Engagement Criteria:

- At least 60% of active users must have more than 5 fanfics saved in their virtual shelves.
- An average of at least 1 written review or rating (stars) generated for every 3 cataloged works.

Performance and Technical Quality Criteria:

- System availability (Uptime) of 99.9%, ensuring users can access their shelves at any time of day.
- Zero critical security failures or user data leaks.